LC Filter Design — Calculations

Design and Harmonic Analysis of a Fault-Tolerant 5-Level CHB Inverter
using LSPWM-PD for EV Traction

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Given Parameters

V_{9a} per cell = 100 V (confirmed in Simulink: v0 = 100 V, two DC sources)
V_{9a} total (2 cells) = 200 V
m_a (modulation index) = 0.95 (sine amplitude = 1.90, carrier peak = 2)
f_{sl} (switching freq) = 4000 Hz (carrier period = 0.00025 s)
f_1 (fundamental) = 50 Hz (sine frequency = 2π × 50)
R (load resistance) = 10 Ω (confirmed in Simulink: R = 10 Ohm)
L_{loax} (load inductance) = 5 mH (confirmed in Simulink: l = 5e-3 H)
Lfl_{l}^{ter} (filter inductor) = 5 mH (confirmed in Simulink: l = 5e-3 H)
Cfl_{l}^{ter} (filter capacitor) = 150 μF (confirmed in Simulink: c = 150e-6 F)
Sample time = 1×10^{-4} s (confirmed in Simulink: SampleTime = 1e-4)

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1. Phase Peak Voltage

Formula:

$$V(peak) = m_a \times (N \times V_{9a})$$

Substitution:

$$\begin{aligned} V(peak) &= 0.95 \times (2 \times 100) \\ &= 0.95 \times 200 \\ &= \mathbf{190\ V} \end{aligned}$$

2. Phase RMS Voltage

Formula:

$$V(rms) = V(peak) / \sqrt{2}$$

Substitution:

$$\begin{aligned} V\ (rms) &= 190 / 1.4142 \\ &= \mathbf{134.4\ V} \end{aligned}$$

Post-filter fundamental from FFT = 200.6 V (peak) ⇒ RMS = 200.6 / √2 = 141.8 V. Slight rise due to +0.67 dB filter gain at 50 Hz.

3. Peak Current

Formula:

$I = V(peak) / R$

Substitution:

$I = 190 / 10$

$= \mathbf{19\ A}$

4. RMS Current

Formula:

$I(rms) = V(rms) / R$

Substitution:

$I(rms) = 134.4 / 10$

$= \mathbf{13.44\ A}$

5. Output Power

Formula:

$P_o = V(rms)^2 / R$

Substitution:

$P_o = (134.4)^2 / 10 = 18{,}063 / 10$

$= \mathbf{1806\ W \approx 1.81\ kW}$

6. Inductor Ripple Current

Formula (20% design rule):

$\Delta I_l = 20\% \times I_{pea}□ = 0.20 \times 19$

$= \mathbf{3.8\ A}$

7. Filter Inductance (L)

Derived from volt-second balance across the inductor in one switching half-cycle:

Formula:

$L = V_{9a} / (4 \times f_{sl} \times \Delta I_l)$

Substitution:

$L = 200 / (4 \times 4000 \times 3.8)$

$= 200 / 60{,}800$

$= \mathbf{3.29\ mH \approx 3.3\ mH}$ **[Calculated value]**

Simulink value used: $L = 5$ mH (confirmed from SLX file: l = 5e-3 H)

5 mH is a standard preferred value larger than 3.3 mH. This gives extra ripple margin (ΔI_l actual ≈ 2.47 A vs. design target 3.8 A) and reduces high-frequency noise further.

8. Filter Cutoff / Resonant Frequency (f^c)

Design rule (target):

$$f^c \text{ (target)} = f_{sl} / 10 = 4000 / 10 = 400 \text{ Hz}$$

Actual value using Simulink components ($L = 5$ mH, $C = 150$ µF):

$$\begin{aligned} f^c &= 1 / (2\pi \times \sqrt{(L \times C)}) \\ &= 1 / (2\pi \times \sqrt{(5\times10^{-3} \times 150\times10^{-6})}) \\ &= 1 / (2\pi \times \sqrt{(7.5\times10^{-7})}) \\ &= 1 / (2\pi \times 8.660\times10^{-4}) \\ &= 1 / 5.441\times10^{-3} \\ &= \mathbf{183.8\ Hz \approx 184\ Hz} \quad \textbf{[Actual Simulink value]} \end{aligned}$$

The report states ~184 Hz — exact match. Lower than the 400 Hz target due to the larger C and L values chosen, which gives stronger attenuation of switching harmonics.

9. Filter Capacitance (C)

Solving for C from the resonant frequency formula, using design target $f^c = 400$ Hz and $L = 3.3$ mH:

Formula:

$$C = 1 / ((2\pi f^c)^2 \times L)$$

Su bstitution (design target):

$$\begin{aligned} C &= 1 / ((2\pi \times 400)^2 \times 3.3\times10^{-3}) \\ &= 1 / ((2513.3)^2 \times 3.3\times10^{-3}) \\ &= 1 / (6{,}316{,}778 \times 3.3\times10^{-3}) \\ &= 1 / 20{,}845 \\ &= \mathbf{48.0\ \mu F} \quad \textbf{[Calculated value]} \end{aligned}$$

Simulink value used: $C = 150$ µF (confirmed from SLX file: c = 150e-6 F)

150 µF is ~3.1× larger than the calculated target. This deliberately lowers f^c from 400 Hz to 184 Hz, providing much stronger attenuation at the switching frequency (extra ~10 dB) at the cost of a slightly slower dynamic response.

10. Characteristic Impedance (Z_0)

Formula:

$$Z_0 = \sqrt{(L / C)}$$

Using design target values (L = 3.3 mH, C = 48 µF):

$$Z_0 = \sqrt{3.3\times10^{-3} \ / \ 48\times10^{-6}} = \sqrt{68.75}$$

$$= \mathbf{8.29\ \Omega} \quad \textbf{[Design target } \mathbf{Z_0}\textbf{]}$$

Using actual Simulink values (L = 5 mH, C = 150 µF):

$$Z_0 = \sqrt{5\times10^{-3} \ / \ 150\times10^{-6}} = \sqrt{33.33}$$

$$= \mathbf{5.77\ \Omega} \quad \textbf{[Actual Simulink } \mathbf{Z_0}\textbf{]}$$

Both values satisfy $Z_0 < R = 10\ \Omega$, ensuring passive load damping in both cases.

11. Quality Factor (Q) and Damping Ratio (ζ)

Using actual Simulink values:

$$Q = R \ / \ Z_0 = 10 \ / \ 5.77$$

$$= \mathbf{1.732}$$

$$\zeta = 1 \ / \ (2Q) = 1 \ / \ (2 \times 1.732)$$

$$= \mathbf{0.289} \quad \textbf{(underdamped — acceptable, load provides passive damping)}$$

12. Damping Resistor Range (R_9)

Formula:

$$R_9 = Z_0 / 3 \quad \text{to} \quad Z_0 / 2$$

Using actual Simulink Z_0 = 5.77 Ω:

$$R_9 = 5.77 / 3 \quad \text{to} \quad 5.77 / 2$$

$$= \mathbf{1.92\ \Omega} \quad \textbf{to} \quad \mathbf{2.89\ \Omega}$$

No separate R_9 is used in the Simulink model. The load resistance ($R = 10\ \Omega$) provides sufficient passive damping since $Z_0 = 5.77\ \Omega < R$.

13. Filter Attenuation at Switching Frequency (4000 Hz)

Frequency ratio:

$$n = f_{sl} \ / \ f^{e} = 4000 \ / \ 183.8 = 21.76$$

Ideal LC attenuation magnitude:

$$A = 1 \ / \ (n^2 - 1) = 1 \ / \ (21.76^2 - 1) = 1 \ / \ 472.5 = 0.002116$$

In decibels:

$$A\ (\text{dB}) = 20 \times \log_{10}(0.002116)$$

$$= \mathbf{-53.5\ dB}$$

This strong attenuation directly explains the THD drop from 30.59% (unfiltered) to 2.24% (filtered) seen in the Simulink FFT results.

14. Fundamental Pass-Band Verification (50 Hz)

Frequency ratio:

$n = f_1 / f^e = 50 / 183.8 = 0.272$

Transfer function magnitude:

$|H| = 1 / |1 - n^2| = 1 / |1 - 0.074| = 1 / 0.926$

$\mathbf{= 1.080 \quad (+0.67\ dB)}$

Verification against Simulink FFT output:

Predicted $Vf_{l}^{l}{}^{tered}$ = 188.5 × 1.080 = 203.6 V

Actual $Vf_{l}^{l}{}^{tered}$ (Simulink FFT) = 200.6 V

Difference = 1.5% (due to load impedance effects in actual simulation)

15. Frequency Modulation Ratio (mf)

Formula:

$mf = f_{sl} / f_1 = 4000 / 50$

$\mathbf{= 80}$

Confirmed in Simulink: carrier period = 0.00025 s ⇒ f_{sl} = 1/0.00025 = 4000 Hz. mf = 80 is an odd multiple, ensuring quarter-wave symmetry and elimination of even-order harmonics.

Comparison: Calculated Values vs. Simulink Values

The table below compares every parameter as calculated using standard design formulas against the actual values extracted directly from the Simulink .slx file (apec_ia2.slx). Where values differ, the reason is explained.

Parameter	Calculated Value	Simulink Value	Verdict
DC voltage per cell (V_{9a})	100 V (given)	100 V (v0 = 100)	**Match**
Total DC bus (2 cells)	200 V	200 V	**Match**
Modulation index (m_a)	0.95 (given)	0.95 (sine amp = 1.9, carrier pk = 2)	**Match**
Switching frequency (f_{sl})	4000 Hz (given)	4000 Hz (T = 0.00025 s)	**Match**
Fundamental frequency (f_1)	50 Hz (given)	50 Hz (2π×50 rad/s)	**Match**
Load resistance (R)	10 Ω (given)	10 Ω (R = 10 Ohm)	**Match**
Load inductance (L_{loax})	5 mH (given)	5 mH (l = 5e-3 H)	**Match**
Peak output voltage	190 V	188.5 V (FFT fundamental)	**Match (~0.8% diff)**

Parameter	Calculated Value	Simulink Value	Verdict
RMS output voltage	134.4 V	141.8 V (post-filter FFT / √2)	**Match (filter gain)**
Peak current	19 A	~18.85 A (derived from FFT)	**Match**
RMS current	13.44 A	~13.4 A (derived from FFT)	**Match**
Output power	1806 W	~1798 W (derived)	**Match**
Ripple current ΔI_l (20%)	3.8 A	~2.47 A (actual, larger L used)	**Conservative**
Filter inductance (L)	3.29 mH (calculated)	5 mH (l = 5e-3 H in SLX)	**Conservative**
Filter capacitance (C)	48 µF (calculated)	150 µF (c = 150e-6 F in SLX)	**Conservative**
Resonant frequency (f^c)	400 Hz (target)	183.8 Hz (actual L & C)	**Intentionally lower**
Characteristic impedance Z_0	8.29 Ω (design target vals)	5.77 Ω (actual vals)	**Both < R = 10 Ω**
Quality factor (Q)	1.205 (design target vals)	1.732 (actual vals)	**Both underdamped**
Damping ratio (ζ)	0.415 (design target vals)	0.289 (actual vals)	**Both underdamped**
Attenuation at f_{sl}	−37.9 dB (design target)	−53.5 dB (actual vals)	**Actual is stronger**
Gain at 50 Hz	+0.17 dB (design target)	+0.67 dB (actual vals)	**Both near unity**
Frequency mod. ratio (mf)	80	80	**Match**
Unfiltered THD (healthy)	Expected high	30.59% (Simulink FFT)	**As expected**
Filtered THD (healthy)	Target < 5%	2.24% (Simulink FFT)	**IEEE-519 PASS**
Filtered THD (faulted)	N/A (fault mode)	42.55% (Simulink FFT)	**Limp-home only**
Sine wave amplitude	1.90 (= m_a × carrier pk)	1.90 (Simulink SLX)	**Match**
Sample time	$1/f_{sl}$ / 10 = 0.0001 s	1e-4 s (Simulink SLX)	**Match**

Explanation of Key Differences

L (3.3 mH calculated vs. 5 mH used):

The calculated value of 3.3 mH satisfies the 20% ripple current design rule. The simulation uses 5 mH — a standard preferred inductor value that provides greater ripple margin and reduces copper losses. The actual ripple current with 5 mH is approximately 2.47 A (vs. 3.8 A design target), well within acceptable limits.

C (48 µF calculated vs. 150 µF used):

The calculated capacitance of 48 µF targets a cutoff of 400 Hz ($f_{sl}/10$). Using 150 µF shifts the resonant frequency down to 184 Hz, more than doubling the separation from 50 Hz and providing approximately 13.6 dB additional attenuation at the switching frequency. This is why the simulation achieves an exceptionally low THD of 2.24% rather than the ~5% that would result from the design-target values.

Resonant frequency (400 Hz target vs. 184 Hz actual):

This is a direct consequence of the larger L and C values. The lower f^c = 184 Hz means the filter operates further into the stop-band at 4000 Hz, giving −53.5 dB attenuation instead of the −37.9 dB that the design-target values would produce. The report confirms ~184 Hz exactly.

Filtered fundamental (190 V calculated vs. 200.6 V in FFT):

The slight rise from 190 V to 200.6 V occurs because the LC filter has a small gain of +0.67 dB at 50 Hz (below resonance). This is expected and harmless. The difference between the calculated 203.6 V and the actual 200.6 V (1.5%) is due to load impedance effects not captured in the ideal LC transfer function.

Summary of Results

V (peak)	= 190 V
V(rms)	= 134.4 V
I	= 19 A
I	= 13.44 A
Output Power (P_o)	= 1806 W
ΔI_l (20% ripple design target)	= 3.8 A
Filter Inductance — calculated	= 3.3 mH
Filter Inductance — Simulink (SLX)	= 5 mH (conservative standard value)
Filter Capacitance — calculated	= 48 µF
Filter Capacitance — Simulink (SLX)	= 150 µF (conservative standard value)
Resonant Frequency — design target	= 400 Hz
Resonant Frequency — Simulink actual	= 183.8 Hz ≈ 184 Hz
Characteristic Impedance (Z_0) — actual	= 5.77 Ω
Quality Factor (Q) — actual	= 1.732
Damping Ratio (ζ) — actual	= 0.289 (underdamped)
Attenuation at f_{sl} = 4000 Hz	= −53.5 dB
Gain at f_1 = 50 Hz	= +0.67 dB

Frequency Modulation Ratio (mf)	= 80
Unfiltered THD — healthy 5-level	= 30.59%
Filtered THD — healthy 5-level	= 2.24% — IEEE-519 PASS (< 5%)
Unfiltered THD — faulted 3-level	= 78.26%
Filtered THD — faulted 3-level	= 42.55% — Limp-home mode